

SLE251 Research Methods and Data Analysis

PRACTICAL WORKSHEET 3: Analysis of variance (ANOVA)

In this practical you will again look, as with the previous practical, at tests for comparing means of different groups. One limitation of t-tests is that they can only compare two groups. Today we will use analysis of variance (ANOVA), which can be used when we want to compare more than two groups. Specifically we will look at a type of ANOVA where there is a single predictor variable/factor/grouping, known as one-factor ANOVA, but also a more complicated version where we can have two predictors (two-factor ANOVA). Fundamentally though they test the same kind of idea as independent samples t-test – is there a significant difference in the mean values for each group being compared.

By the end of this worksheet you should be able to:

- 1) Get R commander to carry out the two different types of ANOVA
- 2) Interpret the results from the above tests in a meaningful manner
- 3) Use Levene's test to test a fundamental assumption of ANOVA, interpret the result of this test and carry out any necessary data transformation

Question 1 – a simple one-factor ANOVA

The Asian citrus psyllid (ACP) is an insect pest which can decimate crops. Ruan et al (2015) were investigating the host-plant preferences of the ACP in the laboratory. They took 6 replicate netted cages into which they placed one seedling each of a citrus, a rhododendron and a plant called Murraya. They then released 60 psyllids into each of the cages. After 4 days they examined the plants and counted the number of ACPs on each plant. They wished to examine whether there was a significant difference in number of ACPs in relation to the identity of plants (i.e. whether there was any consistent preference for which type of plant the ACPs attacked). The data are in the file *sle251psyllid*

Format of sle251psyllid data file

Hostplant	Psyllids
Citrus	20
Citrus	30
""	..
Murraya	23
Murray	16
""	..
Rhododendro	17
n	
""	..

Hostplant The species of plant examined (Citrus, Murraya or Rhododendron)

Psyllids The number of psyllids found on the plant after 4 days

Variables are in columns with levels of the categorical variable listed repeatedly. Each experimental or sampling unit (in this example, each plant) is a separate row.

- a) What is the null hypothesis for this experiment?

b) The appropriate statistical test for comparing the means of more than two groups is an ANOVA *F*-test. What are the assumptions of an ANOVA *F*-test? (to give you two hints: the ways these are checked are listed below, and the assumptions are the same as for independent samples *t*-tests)

Assumption	How is this checked?
1.	Visually using box plots
2.	Visually using box plots, and Levene's test of homogeneity of variances
3.	Thinking about the experimental design and whether true replication has been achieved

These ANOVA assumptions cannot easily be checked using boxplots when sample sizes are small (as is the case with this data set). Ideally boxplots require at least 8 replicates from which to calculate the median and quartiles and provide a realistic description of the shape of the sample. However, it is the second assumption that is of most concern for ANOVA.

Test the null hypothesis that the population group means are equal using a single factor ANOVA. Check the assumption of homogeneity of variances by using Levene's test for homogeneity of variances.

To run a one-way ANOVA in R

Statistics

Variances

Levene's test...

Pick the grouping variable or factor (*Hostplant*) and response variable (*Psyllids*)

Click **Mean** under **Center**

OK

Means

One-way ANOVA...

Enter name for results object in box labeled "Enter name for model:" (note: default name AnovaModel.1 is OK)

Choose *Hostplant* from *Groups* list

Pick *Psyllids* from *Response Variable* list

Check the box for **Pairwise comparison of means**

OK

There will be quite a bit of output to scroll back on in the Output Window – the ANOVA table (under "summary(...)"), followed by the summary stats for each group (mean, sd, n), Tukey's results (under "Simultaneous Tests")

Tukey's results: These are a bit confusing in R. Essentially, it gives the absolute difference (Estimate) and 95% CI for each pairwise difference between means. If the CI (lwr – upr) includes zero, then that pairwise comparison is not significantly different. You can easily visualize this by looking at the **95% family-wise confidence level** plot produced during the analysis. The probability associated with each comparison is also given with a *t*-value. Finally a lettering system is given in the output – each group is assigned a letter, if the letters are different then this indicates significant differences between the groups.

c) What is the null hypothesis for a Levene's test?

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d) What are the conclusions from your Levene's test – are your variances homogeneous or different?

If Levene's test is non-significant it indicates that there no significant difference between group variances.

e) Is there any need for a transformation (log-transformation or otherwise) of these data?

Look at the ANOVA table (you can find the values in your output under "summary(AnovaModel.1)"), and find the F value, the degrees of freedom (two values) and P-value (labeled Pr(>F) in R).

f) Write out the biological outcome as if for a report/thesis, e.g., select the correct statement (delete the wrong statement) and fill in the blanks.

The mean number of Asian citrus psyllids attacking each plant...

was not significantly different

was significantly different

between the three host plant types ($F=$, $df=$, $P=$).

g) What would be the appropriate conclusion to draw from this analysis?

h) Such a result would normally be accompanied by a graph to illustrate the mean (and standard error) number of psyllids on each type of host plant. Construct such a graph. Note: to add the your R plot to the worksheet simply save as a pdf or jpeg and then import it into the space below

To create a plot of means in R
Graphs

Plot of means...

Select *Hostplant* from *Factors* list

Select *Psyllids* from *Response variable* list

Under the *Options* tab Choose *Standard Errors* (under the *Error Bars* heading)

Also under the *Options* tab untick *Connect profiles of means*

OK

i) Whilst we have now established that there is a statistical difference between the group means, we do not yet know which group(s) are different from which other(s). For these data, a Tukey's (post-hoc) multiple comparison test (to determine which host plants are different from each other, in terms of number of psyllids) is appropriate. Complete the following table for Tukey's pairwise comparison of group means (already done as part of the ANOVA above under "Simultaneous tests for general linear hypotheses"): include differences between group means (Estimate) and Tukey adjusted *P*-values (in brackets) for each pairwise comparison (Citrus vs *Murraya* is already filled in).

	<i>Murraya</i>	Citrus
Citrus	-7.333 (0.0307)	
Rhododendron		

j) Likewise in the final part of the output there are letters associated with the groupings (host plant type) which indicate significant differences.

Put the letterings here

Citrus

Murraya

Rhododendron

k) From the plot of means and Tukey's test, what are the correct conclusions about differences in psyllids infesting host plants?

Question 2. A slightly less simple one-factor ANOVA

Lindblad et al (2014) were investigating issues associated with hearing loss in workers from four different professions. Those who work in heavy industry, those in education, those in the music industry and a group of 'others' who are assumed to have minimal exposure to loud noise. They subjected individuals to a 'speech in noise' test, where the individual is played a series of words read by a voice against a loud (>80 decibel) background of white noise. The subject has to write down the words they hear (or think they hear), and the percentage of correct words is noted. Individuals with higher percentages are interpreted to have more acute hearing. The data from the study are in the file *SLE251noise*

Format of sle251noise data file

Work	SN_score
Education	11.7
..	..
Other	44.9
..	..
Music	49.1
..	..
Industry	5.1
..	..

Work The profession that the subject works in

SN_score The result of the 'speech in noise' test (% of words heard correctly)

a) Check the ANOVA assumptions using a boxplot of SN_score against type of work. Any evidence of skewness? Outliers? Does the spread of data look homogeneous between the different work professions?

R Instructions

Graphs

Boxplot

Select *SN_score* from *Variable* list

Click *Plot by groups...* button and choose *Work* and click OK

OK

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Test the null hypothesis that the population group means are equal using a single factor ANOVA and use Levene's test to test for homogeneity of group variances.

To run a one-way ANOVA in R

Statistics

Variances

Levene's test...

Pick Grouping variable (*Work*) and response variable (*SN_score*)

Click **Mean** under **Center**

OK

Means

One-way ANOVA...

Enter name for results object in box labeled "Enter name for model:" (note: default name is OK)

Pick *SN_score* from *Response Variable* list

Choose *Work* from *Groups* list

Check box for Pairwise comparison of means

OK

There will be quite a bit of output to scroll back on in the Output Window – the ANOVA table (under "summary(...)"), followed by the summary stats for each group (mean, sd, n), Tukey's results (under "Simultaneous Tests")

Tukey's results: These are a bit confusing in R. Essentially, it gives the absolute difference (Estimate) and 95% CI for each pairwise difference between means. If the CI (lwr – upr) includes zero, then that pairwise comparison is not significantly different. You can easily visualize this by looking at the **95% family-wise confidence level** plot produced during the analysis. The probability associated with each comparison is also given with a t-value. Finally a lettering system is given in the output – each group is assigned a letter, if the letters are different then this indicates significant differences between the groups.

b) What is the conclusion from Levene's test – variances homogeneous or different? Is there any need for a data transformation?

c) Write out the biological outcome as if for a report/thesis, e.g., select the correct statement (delete the wrong statement) and fill in the blanks.

The mean score in speech to noise tests...

was not significantly different

was significantly different

between the four groups of workers ($F=$, $df=$, , $P=$).

d) What would be the appropriate conclusion to draw from this analysis?

e) Now determine which worker groups are different from each other in terms of hearing acuity, based on Tukey's multiple comparison test (see Question 1 for details). Give the letters associated with each grouping below (1 mark)

Education

Industry

Music

Other

SLE251 Worksheet 3 Analysis of variance

f) Construct such a plot of means, as you did in Question 1, but this time label the means with the letters from the multiple comparison test (see instructions below). Paste your plot of means below

To create a plot of means in R

Graphs

Plot of means...

Select *Work* from *Factors* list

Select *SN_score* from *Response variable* list

Under the *Options* tab Choose *Standard Errors* (under the *Error Bars* heading)

Also under the *Options* tab untick *Connect profiles of means*

OK

To add labels to indicate the results of Tukey test to a plot of means in R

After creating your plot of means

Note down the results of your Tukey test (see answer to e) above) and determine which groups are not significantly different from one another.

In this example we will assume that Education and Industry are one group (we'll call this group 'a') and Music and Other are another (we'll call this group 'b') but your results WILL be different.

Determine what position (left to right) the groups were plotted in your plot of means. You should see they are ordered alphabetically: Education, Industry, Music, Other. Then following this order note down the order that your group labels will be plotted (left to right) so "a" (Education), "a" (Industry), "b" (Music) and "b" (Other).

Select the R script window and type the **bold** text (**NB – due to a quirk with MSWord a copy and paste of this will not work – you will have to type the text below**)

labels <- c('a','a','b','b') The order of the letters (your group labels) should be in the order that you've just determined

Hit the submit button

Select the R script window again

mtext(labels, line=-1, at=1:4) **Note:** The 4 in "at=1:4" represents the total number of treatments in your ANOVA so an ANOVA with only 3 groups would have "at=1:3"

Hit the submit button

The labels should now appear at the top of the plot of means.

h) From the plot of means and Tukey's test, what are the correct conclusions about differences in hearing ability between the four groups of workers?

Question 3 Two factor ANOVA

Martins et al (2015) examined differences in use of male headbobbing displays (used to attract females) in two species of lizard (*Sceloporus parvus*, *Sceloporus cozumelae*), where there are two types of males based on colour patterns on their belly (either colourful, or plain). They wished to examine whether colourful males have different patterns of headbobbing behaviour compared to plain males, and whether the different species have different use of headbobbing behaviour. However, they also wish to test whether the relationships of headbobbing behaviour to colouration differed in the two species (i.e. whether there is an interaction between colouration and species). The appropriate test for this is a two-factor ANOVA.

The data are in the file *sle251lizards*.

Format of sle251lizards data file					
Species	Colour	Headbobs	Group	Species	Species of lizard being observed
parvus	colourful	1	A		
""	""	""	""		
parvus	plain	4	B	Colour	Whether the male had colourful belly patterns or plain
""	""	""	""		
cozumelae	colourful	1	C	Headbobs	The number of Headbobs per minute
""	""	""	""		
cozumelae	plain	3	D	Group	Identifies all 4 groups. We need this in R for Levene's test
""	""	""	""		

a) What null hypotheses are being tested by this two factor ANOVA?

H_0 main effect 1:

H_0 main effect 2:

H_0 interaction:

b) Carry out a Levene's test. What is the result of Levene's test? What does it suggest about the assumption of equality of variances?

To run the Levene's test

Statistics

Variances

Levene's test...

Pick Grouping variable (*Group*) and response variable (*Headbobs*)

Click **Mean** under **Center**

OK

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A log-transformation is not possible on these data as they are (there are zero values in the dataset), consequently we shall try a different data transformation. The type of transformation we use usually depends on the type of data we have – since our data is count data, a square root transformation often works best and so we should try that one first. If our data had been more truly continuous, a log transform is often best (and in instances where we have zeros in our data, we can simply add 1 to all the values – i.e. the transformation is $\log(x + 1)$). Now to our square root transform:

To perform a square root transformation in R

Data

Manage variables

Compute new variable

Enter name for new variable (say *sqrtheadbobs*)

Enter expression to compute - $\sqrt{\text{Headbobs}}$

OK

Note: This will create a new column in the data file with the square root of the number of headbobs which you can then analyse with the ANOVA.

Carry out the Levene's test again using the *sqrtheadbobs* data.

c) what is the conclusion of the Levene's test this time? Does this result suggest that the transformed data would be more appropriate to use in this analysis?

Now carry out the two-factor ANOVA using the appropriate data (*Headbobs* or *sqrtheadbobs*).

To run a two-factor ANOVA in R

Means

Multi-way ANOVA...

Enter name for results object in box labeled "Enter name for model:" (note: default name is OK)

Pick *Headbobs* or *sqrtheadbobs* (as you deem fit) from *Response Variable* list

Highlight *Species* and *Colour* from *Factors* list

OK

Note: To highlight two separate factors: click on one then hold down 'Control' (or command on some, but not all, mac versions – try both!) on the keyboard and click on the second

Output will appear in Output Window – ANOVA table, summary stats for each group (mean, sd, n).

d) complete the table below from the results of the ANOVA.

Source	df	F	P
Colour			
Species			
Colour*Species			
Residual			

d) Draw a plot of means with mean number of headbobs on the y-axis, colour type on the x-axis and the 2 species as different bars. Note: to add the your R plot to the worksheet simply save as a pdf or jpeg and then import it into the space below

(1 mark)

To create a plot of means for different factors in R

Graphs

Plot of means...

SLE251 Worksheet 3 Analysis of variance

Highlight *Species* and *Colour* in *Factors* list
Select *Headbobs* or *sqrtheadbobs* (as you deem appropriate) from *Response variable* list
Under the Options tab Choose *Standard Errors* (under the Error Bars heading)
Also under the Options tab untick Connect profiles of means

OK

e) What conclusions would be most appropriate to draw from the graph and the ANOVA? Remember, it is best to first consider the interaction effect, and assess if it is significant or not – if it is, then this will affect how we interpret the main effects.